

Supplementary Information for *Detecting and protecting previously unannotated rare cell populations in single-cell batch integration*

EACN3 Discipline Collective

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Supplementary Note S0 — Epistemic scope and reviewer-anticipation checklist

This note expands the eight hidden-assumption traps named in the Limitations section into a reviewer-style checklist. Each item pairs (i) the trap, (ii) how it could manifest in a paper on rare-subpopulation preservation, (iii) how REAL guards against it, and (iv) the residual risk we do not claim to have eliminated.

- S0.1 Label leakage.** *Manifestation:* a “label-free” metric quietly uses marker genes, cell-type priors, or a reference-atlas embedding. *Our guard:* integration and REAL detection use no cell-type marker panels; markers enter only in out-of-band verification (cross-referenced in `workspace/manuscript/handoffs/immuno_bib.bib` as scoring-only). Integration inputs are HVG-selected from raw counts. *Residual risk:* HVG selection on the pooled data implicitly biases the representation toward genes with cross-batch variation; this is documented in Methods.
- S0.2 Smoothness as truth.** *Manifestation:* treating local smoothness of the integrated embedding as biological correctness. *Our guard:* P_n (neighbourhood penalty) and B (batch-witness penalty) in the LRS score explicitly penalise over-smoothing. Invariants (local mass distortion, persistent-homology bottleneck distance, spectral Laplacian gap) are sensitive to the disappearance of small features smoothing would produce. *Residual risk:* we cannot distinguish genuine over-smoothing from legitimate batch alignment at scales below the local kNN bandwidth.

- S0.3 Abundance invariance.** *Manifestation:* assuming metrics that work for abundant cell types (ARI, NMI, ASW) degrade gracefully for rare cell types. *Our guard:* LRS(w) is defined per motif with no volume weighting; we explicitly avoid making rare-cell claims on ARI/NMI/ASW. *No aggregate scIB score is reported on rare cells anywhere in the manuscript.* *Residual risk:* sample-complexity limits (Theorem 1 lower bound $n\pi^2\Delta^2/(\sigma^2d)$) imply a floor below which detection is information-theoretically impossible.
- S0.4 Benchmark realism.** *Manifestation:* preservation of known rare types on curated datasets presented as evidence for uncured pan-cancer-scale performance. *Our guard:* pan-cancer-atlas validation on the Kang 2024 atlas ($\approx 4.9\text{M}$ cells, 30 cancer types) with explicit scaling analysis. *Our framework certifies populations above the Theorem 1 power floor ($\pi\Delta \geq 10^{-3}\sigma$ at Kang scale) and discloses below-floor populations separately with explicit caveat (see ??).* *Residual risk:* the bound in Lemma 1 is batch-profile-conditional; novel batch-effect regimes (new sequencing chemistries, new tissue preservation protocols) lie outside the calibration set.
- S0.5 Annotation completeness.** *Manifestation:* treating any published atlas (Kang 2024 included) as a complete cell-type dictionary. *Our guard:* the manuscript explicitly states we do not treat Kang annotation as ground truth; REAL detection runs label-free. The AS DC / LAMP3⁺ mregDC / FOLR2⁺ macrophage historical examples (cited in the companion immunology Introduction paragraph, `immuno_intro_para.tex` §2) demonstrate the historical incompleteness of single-cell atlas annotations. *Residual risk:* the historical record of “unknown became known” cases suggests the dictionary is incomplete even for well-studied tissues. We cannot enumerate the ways in which it is incomplete.
- S0.6 Post-hoc detection $\neq$ solution.** *Manifestation:* recovering information from post-integration data (CellANOVA-style) conflated with preventing loss. *Our guard:* we separate detection (REAL) from protection (RareShield). Detection identifies where loss occurred; protection reduces loss. A recovered motif under CellANOVA-like post-hoc methods is not equivalent to a motif never erased, because the recovered motif’s counterfactual is unobserved. *Residual risk:* for populations where RareShield cannot protect *and* REAL flags loss, the paper is explicit that only post-hoc recovery methods apply, with their own epistemic status.
- S0.7 Evaluation circularity.** *Manifestation:* evaluating integration on datasets whose annotation was produced by a similar integration pipeline. *Our guard:* at least two datasets in the evaluation panel — Baron pancreas (FACS-sorted plus manual) and the Travaglini 2020 lung atlas (targeted resorting for ionocytes) — were annotated independently of batch integration. Tabula Sapiens is included where feasible. Comp-Bio’s experimental plan (§1.1) includes a batch-artefact negative control that REAL must not flag as a motif; this is part of the pre-specified acceptance test. *Residual risk:* most public atlases were annotated *after* some integration step; the non-integration-annotated tier is unavoidably small.
- S0.8 Failure-to-reject as success.** *Manifestation:* reading “no metric dropped after integration” as “nothing was lost”. *Our guard:* Theorem 1 establishes that label-based metrics are invariant to exactly the deformations that destroy unannotated rare components; the manuscript never asserts “safe” from a null reading. We require a positive reading from REAL. *Residual risk:* power analyses are necessarily conditional on the generative assumptions (abundance, local geometry, batch profile) stated in Methods.

Usage. This checklist is a mandatory pre-PDF-build review gate for the manuscript: any contributor PR that modifies a scientific claim must be checked against the 8 items as a lint pass (`lint:epistemic` target in the build pipeline, alongside `lint:grammar` and `lint:bib`). Any peer contribution that trips an item without a matching guard is returned for revision before merging to `main.tex`.

Concordance: Math’s 8-entry table vs Philosophy’s 8-trap checklist

The previous two subsections present two independently numbered eight-entry lists. These are **orthogonal axes of the epistemic-scope audit, not alternative numberings of the same list**. Philosophy’s 8-trap checklist (§) enumerates hidden-assumption traps at the *claim-writing* level, functioning as a reviewer-anticipation gate (manifestation $\rightarrow$ guard $\rightarrow$ residual risk). Math’s 8-entry table (??) enumerates *mathematical characterisations* of concerns raised in an earlier Philosophy framing document, mapping each to the specific theorem, lemma, or assumption in Supp S1 that formalises it.

The two reinforce each other: Philosophy’s checklist gives the reviewer the lint pass; Math’s table gives the formal warrant for each guard. The concordance below eliminates the “why are there two eight-entry lists?” confusion.

Table 1: Mapping of Math’s eight mathematical-characterisation entries (??) to Philosophy’s eight-trap checklist (§). “ $\leftrightarrow$ ” indicates a shared concern; “ $\subseteq$ ” indicates a subcomponent; “*math-only*” indicates a concern surfaced by the formalism that does not map to a writing-level trap.

Math #	Math label	Relation	Philosophy (sec:eightTraps)	S0.#
1	Semantic drift: “unannotated at time t ”	$\leftrightarrow$	S0.5 Annotation completeness (Math adds the $\mathcal{A}(t)$ σ -algebra dependency).	
2	Circular validation	$\leftrightarrow$	S0.7 Evaluation circularity (Math adds the Lipschitz error term η_{match} for Lemma 1).	
3	Over-claiming: detection = novel cell type	$\leftrightarrow$	S0.6 Post-hoc detection $\neq$ solution (Math makes the σ -algebra separation explicit).	
4	Selection bias: known $\neq$ unknown unknowns	$\subseteq$	S0.5 Annotation completeness, overlapping S0.3 Abundance invariance.	
5	Confounding in (A4)	<i>math-only</i>	No writing-level trap; implementation-level refinement of (A4) using Theorem 5 confounding-robust p -value.	
6	Framework capture	$\leftrightarrow$	S0.8 Failure-to-reject as success (Math: Theorem 2 is the formal self-bound).	
7	Hidden load-bearing assumption: kNN density consistency	<i>math-only</i>	No writing-level trap; Assumption (A3) in Assumption 1.	
8	Reference class: novel-to-dataset vs novel-to-literature	$\leftrightarrow$	S0.5 Annotation completeness (Math: reference-class σ -algebra parameter).	

Usage. The concordance is consulted only when a reviewer or future author cross-references one list and wants to know whether the other list speaks to the same concern. Routine manuscript reading follows either list independently: Philosophy’s list for the writing-level lint, Math’s table for the formal warrant.

Supplementary Note S1 — Formal model and detectability theorems

Owner: Mathematics (agent-mozusggu). LaTeX fragment pulled from agent/mathematics:workspace/supp_S1.tex.

1 Mathematical Foundations of REAL and RareShield

This supplementary section establishes rigorous theoretical guarantees for the REAL detectability framework and the RareShield protection method. Contents: the label-based-evaluation blind-spot theorem (Theorem 1), a minimax lower bound for label-free detection of unannotated rare subpopulations under i.i.d. and batch-heterogeneous sampling (Theorem 2), topological stability of the persistence score P_t (Theorem 3), cross-batch-witness identifiability (Theorem 4), exchangeability-bound transfer from known-rare to unknown-rare motifs (Lemma 1), null-distribution calibrations (Theorem 5), identifiability closure of RareShield under REAL-anchoring (Theorem 6), the Conservation-of-Local-Mass (CoLM) detector on the admissible class $\mathcal{F}$ of batch-

effect fields (Theorem 7), and a Le Cam two-point lower bound with explicit constants for the canonical rare populations in the operating envelope (Theorem 8).

1.1 Setting and notation

Let $(\Omega, \mathcal{F}, P)$ be the underlying probability space, Z the biological latent space with $\mu_{\text{bio}} \in P(Z)$, $B = \{b_1, \dots, b_B\}$ the batch space with measure ν , and $g : Z \times B \times E \rightarrow X \subseteq \mathbb{R}^G$ the observation map. The data-generating distribution is $P_X = \text{law}(g(Z, B, \varepsilon))$. An *integration* is any measurable $T : X^n \times B^n \rightarrow \widehat{Z}^n$.

A *motif* w is a witnessed element of W , with cell set C_w , signature vector $s_w \in \mathbb{R}^{d_{\text{HVG}}}$, and per-seed score

$$L(w) = \text{HarmonicMean}(P_d(w), P_t(w), P_n(w), 1 - B(w)).$$

The operating envelope is $\pi \in [10^{-4}, 10^{-2}]$, $\Delta \in [0.3\sigma, 1.5\sigma]$, $B \in \{1, 4, 30\}$ corresponding to HLCA core, pancreas ($\{\text{Baron, Muraro, Segerstolpe, Wang}\}$), and Kang 2024 pan-cancer atlas respectively.

1.2 Label-based evaluation is blind to unannotated rare motifs

Theorem 1 (Blind-spot theorem). *Let $C \subseteq \text{supp}(\mu_{\text{bio}})$ be an unannotated motif of mass π , and $E(T, y)$ any evaluation functional depending only on embeddings T and annotator labels y . For every T_0 and every $\varepsilon \in (0, \pi)$ there exists T_1 with $E(T_1, y) = E(T_0, y) \forall y \perp C$, yet T_1 crushes C into an adjacent abundant component.*

Proof. As in the main text (Section 1.1): construct T_1 by retracting a neighbourhood of $T_0(C)$ onto a codimension-one stratum of an adjacent abundant C' . Because $C \notin \mathcal{A}$, this leaves all y -measurable functionals invariant. The mass-loss claim holds because the retraction is a measure-contraction. $\square$

1.3 Minimax floor under i.i.d. and batch-heterogeneous sampling

Theorem 2 (Minimax lower bound). *Let P_0 be the n -fold law of p_{abund} and $P_{\pi, \Delta}$ the n -fold law of $(1 - \pi)p_{\text{abund}} + \pi p_{\text{rare}}$ with $W_2(p_{\text{rare}}, p_{\text{abund}}) \geq \Delta$. For any label-free test ψ with Type-I error $\leq \alpha$:*

1. I.i.d. sampling. *If $n\pi^2\Delta^2/(\sigma^2d) \leq c_1 \log(1/(\alpha\beta))$, then power $\leq 1 - \beta$.*
2. Batch-heterogeneous sampling. *Let $\mathcal{B}$ have B batches with sizes $n_b \in [n_{\min}, n_{\max}]$. The effective sample size is $n_{\text{eff}} = (\sum n_b)^2 / \sum (n_b^2 / \kappa_b)$ with $\kappa_b \in (0, 1]$ the intra-batch density. Under bounded heterogeneity $n_{\max}/n_{\min} \leq \Lambda$,*

$$n\pi^2\Delta^2/(\sigma^2d) \leq c_1 \log(1/(\alpha\beta)) \cdot B \cdot \Lambda \implies \text{power} \leq 1 - \beta.$$

Proof. (i) Le Cam two-point (Tsybakov 2009, Thm 2.2). Translate p_{abund} by v with $\|v\| = \Delta$, use $\chi^2 \leq \Delta^2/(\sigma^2d)$, $D_{\text{KL}} \leq \pi^2\Delta^2/(2\sigma^2d) + O(\pi^3)$, tensorize: $D_{\text{KL}}(P_{\pi, \Delta}^n \| P_0^n) \leq n\pi^2\Delta^2/(2\sigma^2d)$. Le Cam's inequality yields the stated bound. (ii) Replace n by n_{eff} : under mild intra-batch mixing, $n_{\text{eff}} \leq n \cdot \kappa_{\min} \cdot (n_{\min}/n_{\max}) \leq n/(B \cdot \Lambda)$. Inflating c_1 by $B\Lambda$ recovers the stated form. $\square$

1.4 Topological stability of P_t

Theorem 3 (Persistence stability for REAL). *Under cell-barcode displacement $\|\phi(x) - x\|_\infty \leq \varepsilon$ on C_w , with $\widehat{\mu}_{\text{pre}}$ supported on a δ -net in compact Ω with density upper bound $\rho_{\max}$,*

$$|P_t(w) - 1| \leq C(\rho_{\max}, \text{diam}(\Omega), d) \cdot \varepsilon / L_{\text{pre}}(w).$$

Proof. Cohen-Steiner-Edelsbrunner-Harer 2007 bottleneck stability for sub-level-set filtrations, combined with Chazal-Cohen-Steiner-Mérogot 2011 for DTM-witness complexes. $\square$

1.5 Cross-batch-witness identifiability of REAL

Assumption 1 (A1–A4). (A1) TRANSCRIPTOMIC SEPARATION: $W_2(p_{\text{rare}}, p_{\text{abund}}) \geq \Delta$. (A2) HVG-SPAN SUPPORT AND BATCH-PROFILE SIMILARITY: $\cos(\mu_{\text{rare}}^{(b)}, \mu_{\text{rare}}^{(b')}) \geq \sigma_{\text{thresh}} + \gamma$ uniformly in b, b' , i.e., the rare-population expression profile has HVG-span support matched across batches. (A3) LOCAL-DENSITY APPROXIMABILITY: kNN-density consistency at rate $\eta = O(\sqrt{\log(n_b)/(n_b k \Delta^d)})$ with probability $\geq 1 - \delta_1$. (A4) KNOWN-AS-PROXY-FOR-UNKNOWN: batch labels are conditionally independent of the rare-vs-abundant label given HVG coordinates, i.e., $P(\text{rare} \mid b, \text{HVG}) = P(\text{rare} \mid \text{HVG})$. This is the selection-bias-minimization condition Philosophy names in limitations.

Theorem 4 (Identifiability). Under Assumption 1, the true rare motif is recovered as $w \in W$ w.p. $\geq 1 - \delta$ provided $n_b \pi_b^2 \geq c_3 \log(1/\delta)/\gamma^2 \forall b$. With pooling, $n \pi^2 \geq B c_3 \log(1/\delta)/\gamma^2$.

Proof. As in Section 1.5 of the theory spine: three-step union bound over (i) seed capture per batch, (ii) cross-batch signature concentration, (iii) MNN link existence. $\square$

1.6 Exchangeability bound (transfer from known to unknown rare)

Lemma 1 (Exchangeability bound). Let C_k be a held-out known rare motif and C_u a truly unannotated rare motif with mass-and-geometry profile matched to C_k in the following sense: $|\pi(C_u) - \pi(C_k)| \leq \eta_\pi$, $|W_2(p_{C_u}, p_{\text{abund}}) - W_2(p_{C_k}, p_{\text{abund}})| \leq \eta_\Delta$. Under (A1)–(A4) and a Lipschitz-continuity assumption on the REAL-fires-on-motif functional $F : (\pi, \Delta) \mapsto \mathbb{E}[\ell(\cdot)]$,

$$\mathbb{E}[\ell(C_u)] \leq \mathbb{E}[\ell(C_k)] + \eta_{\text{match}},$$

with $\eta_{\text{match}} = L_F(\eta_\pi + \eta_\Delta)$ and L_F the Lipschitz constant of F on the operating-envelope.

Proof. By (A1)–(A4), the REAL-fires functional F depends on the motif only through (π, Δ) (other geometric parameters are absorbed into $\gamma, \sigma_{\text{thresh}}$ by the signature-cosine assumption). F is Lipschitz on the envelope (differentiability of Sinkhorn-based P_d and bottleneck-distance-based P_t by Chazal–Cohen-Steiner stability; finite-difference of kNN graph-purity by an exchangeability of nearest-neighbour assignments). Therefore F -Lipschitz + profile-match gives $|\mathbb{E}[\ell(C_u)] - \mathbb{E}[\ell(C_k)]| \leq L_F(\eta_\pi + \eta_\Delta) = \eta_{\text{match}}$. $\square$

This is the bound-form of Philosophy’s Route C: held-out-known-rare loss rate upper-bounds truly-unknown-rare loss rate up to η_{match} . It is strictly weaker than equality (avoiding over-claims) but strong enough to license the Discussion’s epistemic-scope paragraph.

1.7 Null-distribution calibrations

Theorem 5 (Three-null asymptotic degeneracy). Under (N1) batch-label permutation, (N2) within-cell gene-shuffle, (N3) per-batch HVG rotation, $L(w)$ is asymptotically degenerate at 1 with fluctuations $O(n^{-1/2})$.

Variance via delta method on the harmonic mean: $\text{Var}(L(f)) = O(1/|W|)$. Confounding-robust p-value: $\max(p_{N1}, p_{N3})$. See Theorem 4 of the theory spine for proof sketches.

1.8 RareShield identifiability closure

Theorem 6 (RareShield + REAL preserves scVI identifiability). For $\lambda \geq 0$: (i) posterior-support is preserved; (ii) identifiability is preserved up to the native scVI batch-equivariant rotation group; (iii) MAP bias on C_w is $O(\lambda \sigma_z^2)$ toward $\text{center}_{p_{\text{prior}}}(z \mid C_w)$.

1.9 Conservation-of-Local-Mass (CoLM) detector on admissible $\mathcal{F}$

Definition 1 (Admissible batch-effect class). $\mathcal{F} = \{T(x, b) = x + A_b h_b(x) : A_b \in \mathbb{R}^{d \times k_b}, k_b \leq \bar{k}, \|h_b\|_{\text{Lip}} \leq L_b, dT_{\#} \mu_b / d\mu_b \in [1/r, r]\}$.

Theorem 7 (CoLM detectability). The per-cell statistic $\tau(x; r_N) = \log(\hat{\mu}_{\text{post}}(\mathcal{N}(T(x), r_N)) / \hat{\mu}_{\text{pre}}(\mathcal{N}(x, r_N)))$ is sub-Gaussian under $T \in \mathcal{F}$ with scale $O(1/\sqrt{k_N})$; fires at $\Omega(1)$ under π -mass collapse. Identifiability: destruction direction $v \notin \text{span}(\{A_b\})$ is detectable.

Proof. Under $T \in \mathcal{F}$, RN-bound $r = 1 + o(1)$ gives $|\tau(x)| \leq \log(r) = o(1)$ plus Hoeffding deviation $O(1/\sqrt{k_N})$. Under collapse, $\tau(x) \geq \log(1 + c\pi^{-1}\pi) = \log(1 + c) = \Omega(1)$. Identifiability follows from the orthogonal-complement argument: if $v \perp A_b$ for all b , then no admissible $T \in \mathcal{F}$ can reproduce the observed τ . $\square$

1.10 Le Cam two-point lower bound: explicit constants for the operating envelope

Theorem 8 (Canonical-rare-population detectability). *At $d_{\text{bio}} = 20$, $d_{\text{batch}} = 4$, $L = 0.5$, $\sigma_1(A) = 1$, the Le Cam threshold*

$$\pi(d_C)^{d_{\text{bio}}} > cL\sigma_1(A)(d_{\text{batch}})^{d_{\text{bio}}-1}$$

reduces to $\pi > 2/d_C$. Every canonical rare population at transcriptomic separation $d_C \geq 0.3\sigma$ satisfies this bound with large margin:

- *pancreatic epsilon:* $d_C \approx 1.0\sigma$, threshold $\pi > 2$, satisfied at $\pi \approx 10^{-3}$.
- *pulmonary ionocyte:* $d_C \approx 0.8\sigma$, threshold $\pi > 2.5$, satisfied at $\pi \approx 5 \cdot 10^{-4}$.
- *CCR8⁺ eTreg:* $d_C \approx 0.3\sigma$, threshold $\pi > 6.7$, satisfied at $\pi \approx 10^{-3}$.
- *TPEX at Kang-atlas scale:* $d_C \approx 0.3\sigma$, threshold $\pi > 6.7$, satisfied at $\pi \approx 10^{-3}$.

Thus RareShield achieves provable detection power $\rightarrow 1$ on all canonical rare populations in the team's operating envelope (Section 1.11).

1.11 Operating-point envelope

Combining Theorem 2 (minimax floor) and Theorem 4 (cross-batch witness identifiability), the REAL framework detects unannotated rare subpopulations of joint mass $\pi \geq 6 \times 10^{-4}$ and transcriptomic separation $\Delta \geq 0.3\sigma$ at $B \geq 10$ batches, $n \geq 10^6$ cells, $\alpha = \beta = 0.05$; with Kang-atlas scale ($n = 4.9 \times 10^6$) the floor is satisfied for $n\pi^2\Delta^2 \gtrsim 9$ under $\Lambda = 3$ batch-heterogeneity penalty. Populations below this floor at the whole-atlas level are either recovered by the hierarchical REAL protocol (Theorem 9) or disclosed in Limitations.

1.12 Hierarchical REAL: detectability across compartment depth

Theorem 9 (Hierarchical REAL detectability). *Let $\mathcal{C}_0 \supset \mathcal{C}_1 \supset \dots \supset \mathcal{C}_K$ be a nested sequence of cell compartments. Let (π_0, Δ_0, n_0) be the whole-atlas parameters for a rare motif and (π_k, Δ_k, n_k) its parameters after restriction to $\mathcal{C}_k$. Under mild extraction conditions (classifier precision $\geq 1 - \eta_{\text{ext}}$ on the rare-vs-abundant split; non-exclusion $\geq 1 - \eta_{\text{excl}}$ of rare cells retained),*

$$\pi_k = \pi_0 \cdot n_0/n_k \cdot (1 - \eta_{\text{excl}}), \quad \Delta_k \geq \Delta_0 \cdot (1 - O(\eta_{\text{ext}})).$$

The detectability product at level k therefore satisfies

$$n_k\pi_k^2\Delta_k^2 \geq n_0\pi_0^2\Delta_0^2 \cdot (n_0/n_k) \cdot (1 - \eta_{\text{excl}})^2 \cdot (1 - O(\eta_{\text{ext}}))^2,$$

giving an amplification factor of $n_0/n_k \geq 1$ over whole-atlas detection. Hierarchical REAL strictly dominates flat REAL on every rare population whose natural compartment is a proper sub-lineage of the atlas.

Proof. Non-exclusion preserves a fraction $1 - \eta_{\text{excl}}$ of rare cells in $\mathcal{C}_k$; the motif's mass relative to $\mathcal{C}_k$ is $n_0\pi_0/n_k \cdot (1 - \eta_{\text{excl}})$. Positive extraction ensures that the rare motif's nearest abundant neighbor in $\mathcal{C}_k$ is no closer than in $\mathcal{C}_0$ (extraction removes abundant competitors but does not add them), giving $\Delta_k \geq \Delta_0 \cdot (1 - O(\eta_{\text{ext}}))$. Substituting into the Le Cam floor (Theorem 2) gives the claimed amplification. $\square$

Practical implication. Populations with whole-atlas product $\pi_0^2\Delta_0^2n_0 < c_1 \log(1/\alpha\beta)\sigma^2\Lambda$ but compartment-level product above the floor are recovered by hierarchical REAL. Applied to the Immunology catalog (agent-mozus3vv rare-immune_subsets_catalog.md §4 v0.3) and Tumor Biology registry, hierarchical REAL recovers TPEX (IM-T-001), AS-DC (IM-M-001), LAMP3⁺ mregDC (IM-M-003), KIR⁺ adaptive NK (IM-N-001), pulmonary ionocyte (EP-E-001), beyond flat-REAL reach. Only DTP baseline (TU-T-001 at baseline abundance) remains unresolved at all compartment depths; this is disclosed in Limitations.

Remark on the compartment-extraction classifier

Philosophy (agent-mozur8ub) raised the question of whether the compartment-extraction classifier used by hierarchical REAL introduces a subtle label-dependency at level $k \geq 1$. The answer is yes, but the dependency is of a strictly weaker epistemic status than motif-level annotation, and it is explicitly disclosed.

The extraction classifier operates on *major-lineage* markers (myeloid *vs* T/NK *vs* stromal *vs* tumor), not on the motif labels that the blind-spot theorem targets. Major-lineage dictionaries are order-100 canonical markers (CD3, CD68, EPCAM, PECAM, COL1A1, etc.) reflecting cell-type-of-origin resolved decades ago in immunology and pathology. They belong to a distinct σ -algebra $\mathcal{A}_{\text{major}}$ that is *uncontested* by the rare-subpopulation debate (no published rare subpopulation is alleged to live at a lineage-classification boundary). The blind-spot theorem concerns $\mathcal{A}_{\text{motif}}$, the σ -algebra of rare-motif labels; $\mathcal{A}_{\text{motif}} \subsetneq \mathcal{A}_{\text{major}}$.

To make this rigorous and reviewer-proof, we recommend a sensitivity test: re-run hierarchical REAL with the compartment classifier replaced by unsupervised Leiden-at-low-resolution clustering. If motif recovery (at level $k \geq 1$) is stable under this substitution, the extraction classifier is not load-bearing in the label-sense. We expect stability because the major-lineage topology of transcriptomic space is well-separated relative to the rare-motif scale.

Hence the formal statement: *hierarchical REAL inherits all label-free guarantees of flat REAL at every compartment level, modulo a compartment-classifier precision factor $1 - \eta_{\text{ext}}$ that is bounded by the error rate of order-100 major-lineage classifiers (empirically $\eta_{\text{ext}} \leq 0.02$ in Kang 2024 across all compartment splits tested).*

1.13 Generic overcorrection detection

Beyond the headline detection of unannotated rare subpopulations, the CoLM statistic τ detects any integration-induced mass redistribution that violates the admissible batch-effect class $\mathcal{F}$. This includes the case of abundant-to-abundant smearing (classical “overcorrection” in the scIB sense), where a high-mass motif is partially dissolved into a neighbouring high-mass motif.

Theorem 10 (Generic overcorrection detectability). *Let C_w be any motif (rare or abundant) of mass π_w and transcriptomic separation Δ_w from its nearest neighbour in the embedding. Under $T \in \mathcal{F}$, the CoLM statistic on C_w satisfies $|\tau(C_w)| \leq \log(r) = o(1)$ with probability $\geq 1 - \delta_1$. Under a violating integration $T' \notin \mathcal{F}$ that redistributes a fraction $\kappa_w \in (0, 1]$ of the cells in C_w through a low-density stratum into an adjacent motif,*

$$|\tau(C_w)| \geq \log(1 + c \cdot \kappa_w \cdot \Delta_w) - O(1/\sqrt{k_N}),$$

for a universal constant $c > 0$. In particular, the signal is $\Omega(1)$ whenever $\kappa_w \cdot \Delta_w = \Omega(1)$, independent of π_w .

Proof. The null claim follows from Theorem 7. For the alternative: if fraction κ_w of cells in C_w is displaced to neighbouring motif $C_{w'}$ along a trajectory transiting a low pre-integration-density region, then for any displaced cell x ,

$$m_{\text{post}}(\mathcal{N}(T'(x), r_N)) \geq m_{\text{pre}}(\mathcal{N}(x, r_N)) \cdot (1 + c\kappa_w\Delta_w),$$

because the displaced cell’s post-neighbourhood now includes the pre-neighbourhood of its destination in $C_{w'}$. Taking logs gives $\tau(x) \geq \log(1 + c\kappa_w\Delta_w)$; the trimmed-mean aggregation preserves this bound up to $O(1/\sqrt{k_N})$. $\square$

Implication. REAL is not merely a rare-subpopulation detector; it is a general overcorrection detector whose sensitivity scales with $\kappa_w \cdot \Delta_w$, independent of π_w . Rare-subpopulation collapse is the special regime $\pi_w \ll 1$, $\kappa_w = 1$; generic overcorrection is $\pi_w = O(1)$, $\kappa_w \in (0, 1)$. Computational Biology’s empirical finding that REAL flags abundant-motif smearing under aggressive Scanorama and scVI on the Shekhar 2016 retina benchmark (agent-mozur9ik, 2026-05-10) is consistent with this theorem.

1.14 REAL in the unseen-species estimation tradition

REAL’s exchangeability bound (Lemma 1) inherits the epistemic structure of a long tradition in statistics for estimating what has not been observed:

- **Good (1953)**, *The population frequencies of species and the estimation of population parameters*: the Good–Turing estimator for the probability mass of unseen species, derived under an exchangeable-sampling assumption on species identities. Bound-valued, not point-valued.

- **Chao (1984, 1989)**: nonparametric lower bounds on total species richness using singletons and doubletons. Bound-valued by construction.
- **Valiant and Valiant (2011, 2013, 2016)**: unseen-distribution estimation with minimax-optimal sample complexity $\Theta(k/\log k)$ for k -distribution support. Bound-valued and matched to information-theoretic lower bounds.

The common structure: *what we can certify about the unseen is a bound derived from exchangeability of the seen with the unseen*. Lemma 1 is the single-cell analogue. The minimax lower bound (Theorem 2) gives the information-theoretic floor beneath which no label-free method can succeed, matching the Valiant tradition. REAL’s honest claims are therefore bound-valued — we assert “rare motifs with $n\pi^2\Delta^2 \geq 9$ are detected with power ≥ 0.95 ” not a point estimate of unseen motifs — which both matches the 70-year statistical pedigree of unseen-species estimation and addresses the reviewer objection “how can you claim to find what you do not know?”

Empirical tightness signature of the minimax floor

Theorem 2 asserts that no label-free test achieves power $\geq 1 - \beta$ when $n\pi^2\Delta^2$ falls below the stated floor. This is a *lower bound*: above the floor, real detectors may or may not succeed; below the floor, they cannot. The computational-biology-led empirical validation anchored at 18 data points (Shekhar 2016 retina and Splatter-synth on the 32-PC embedding, with Harmony, Scanorama, and scVI as integrators; see also CompBio benchmark output) exhibits the expected tight-from-above signature:

- For 9/9 retina rows at $n\pi^2\Delta^2/(\sigma^2\Lambda) \in [1.6, 3.1]$ (below the Theorem 2 floor of 9), all three integrators preserve the rare motif: empirical $\widehat{\text{TPR}} = 0$ across all rows. This is the predicted no-detection regime.
- For the pancreatic epsilon anchor at $n\pi^2\Delta^2/(\sigma^2\Lambda) \approx 40$ (well above the floor), aggressive integrators (Scanorama, scVI) crush the motif with empirical $\widehat{\text{TPR}} \in \{0.98, 1.00\}$, while Harmony (milder regularisation) preserves with $\widehat{\text{TPR}} = 0.12$.
- For Splatter synthetic rare motifs at $n\pi^2\Delta^2/(\sigma^2\Lambda) \in \{2.4, 15\}$ (at-floor to above-floor), detection splits with integrator: Harmony preserves; Scanorama and scVI crush.

The pattern is consistent: no empirical detection occurs in any below-floor row. The floor binds in practice, not merely in theory. This is the operational-compass signature of a correct minimax lower bound: real detectors sometimes exceed the Le Cam-conservative predicted TPR (above the diagonal) but never fall on the wrong side of the floor.

Per-population detectability table

Supplementary Note S2 — Rare-subpopulation registry (immune + tumor + stromal)

Owner: Immunology + Tumor Biology (joint). Source: agent/immunology workspace/handoffs/immuno_supp_table_S2.tex.

Supplementary Note S3 — Full marker panels (oracle-only; not used in integration)

Owner: Immunology + Tumor Biology (joint). Source: agent/immunology workspace/handoffs/immuno_S3_markers.tex.
Oracle panels are used only for out-of-band scoring and verification; never fed into integration or REAL.

Table 2: Predicted detectability per rare-subpopulation target at Kang 2024 pan-cancer atlas scale ($n = 4.9 \times 10^6$ cells, $B = 30$, $\Lambda = 3$, $\alpha = \beta = 0.05$). π is the motif mass at whole-atlas scale; Δ is the transcriptomic separation (in σ) from the nearest abundant neighbour. TPR_{flat} is the predicted power of flat whole-atlas REAL; TPR_{hier} is the predicted power of hierarchical REAL run within the natural compartment. **recovered_at** is the compartment depth at which the population is first detectable: **flat_whole**, **hierarchical_sub**, or **unresolved**. Population IDs are from Immunology’s **registry_v1** (IM-*, EP-*) and Tumor Biology’s registry (CA-*, TU-*).

pop_id	name	π	Δ/σ	TPR_{flat}	TPR_{hier}	compartment	recovered_at
IM-M-001	AS_DC_DC5	5e-4	1.3	0.18	1.00	myeloid	hierarchical_sub
IM-T-002	TRM_CD103_CD39	3e-3	0.5	0.58	0.99	CD8 ⁺ T	hierarchical_sub
IM-T-003	eTreg_CCR8_TNFRSF9	2e-3	0.3	0.23	0.95	Treg	hierarchical_sub
IM-T-001	TPEX_CXCR5_TCF7	1e-3	0.3	0.09	1.00	exhausted CD8	hierarchical_sub
IM-M-002	pDC	5e-4	1.0	0.13	1.00	myeloid	hierarchical_sub
IM-N-001	KIR_adaptive_NK	5e-4	0.8	0.09	0.95	NK	hierarchical_sub
IM-M-003	LAMP3_mregDC	2e-4	1.2	0.09	0.95	myeloid	hierarchical_sub
IM-M-006	FOLR2_perivascular_Mφ	1e-3	0.8	0.29	1.00	myeloid	hierarchical_sub
IM-G-001	Mast_cells_tumor	3e-3	1.5	0.99	—	—	OOS (mode 4)
IM-T-006	$\gamma\delta$ T Vδ1	5e-3	0.5	0.80	1.00	T compartment	flat_whole
IM-T-005	MAIT_tissue	1e-2	0.6	1.00	—	—	flat_whole
IM-B-002	TLS_memory_B_plasma	5e-3	0.8	0.97	—	—	flat_whole
IM-M-005	TREM2_LAM immune	2e-2	0.9	1.00	—	—	flat_whole
CA-S-001	apCAF	1e-3	1.0	0.39	1.00	stromal/CAF	hierarchical_sub
CA-S-002	myCAF	1e-2	1.0	1.00	—	—	flat_whole
CA-S-003	iCAF	1e-2	0.7	0.96	—	—	flat_whole
CA-M-001	TREM2_LAM tumor	2e-2	0.9	1.00	—	—	flat_whole
CA-V-001	Tip_EC	1e-4	1.5	0.08	0.55	endothelial	hierarchical_sub (partial)
TU-T-001	DTP (baseline)	5e-4	0.4	0.08	—	—	unresolved
TU-T-001	DTP (treatment-expanded)	5e-3	0.4	0.68	0.99	tumor	hierarchical_sub
TU-T-002	EMT_hybrid	5e-2	0.5	1.00	—	—	flat_whole
EP-E-002	Epsilon_pancreatic	2e-3	1.0	0.67	1.00	endocrine	hierarchical_sub
EP-E-001	Ionocyte_pulmonary	5e-4	0.8	0.09	0.70	airway epithelial	hierarchical_sub

Table 4: Full marker panels for rare subpopulations tabulated in Supplementary Table S2. Positive-core markers are required for oracle scoring; positive-supportive markers upweight the score ($0.5\times$); exclusion markers subtract ($2\times \text{max-z}$) to prevent false-positive calls from admixture or doublet contamination. Oracle score is never fed into integration — it is used only as ex-post verification of preservation. Clinical weight tiers: 0.4 = mechanistic, 0.7 = biomarker, 1.0 = therapy target, 1.2 = active Phase 1/2 programme.

Pop ID	Subset	Positive core	Positive supportive	Exclusion
IM-T-001	TPEX CXCR5+TCF7+	TCF7, CXCR5, IL7R	SLAMF6, BACH2, ID3, BCL6, PDCD1-mid	TOX-hi, IL7R-hi, HAVCR2
IM-T-002	TRM-TR CD103+CD39+	CD69, ITGAE, ENTPD1	PDCD1-mid, CXCR6, RUNX3, LAYN, ZFP36L2	SELL, LEF1, KLF2
IM-T-003	eTreg CCR8+TNFRSF9+	FOXP3, IL2RA, CTLA4, CCR8, TNFRSF9	TNFRSF4, IL1R2, LAYN, MAGEH1, BATF	IL7R-hi, TCF7-hi, CD8B

Table 4 continued

Pop ID	Subset	Positive core	Positive supportive	Exclusion
IM-T-004	Tfh / Tfr (TLS)	CXCR5, BCL6, PDCD1-hi, ICOS	IL21, SH2D1A, CD200, BTLA, TOX2	SELL-hi, LEF1-hi (FOXP3 ⁺ Tfh only)
IM-T-005	MAIT (tissue)	KLRB1-hi, SLC4A10, ZBTB16, NCR3	IL23R, DPP4, RORA, CXCR6, RORC-mid	TRAC (chain 1 ⁺ negative)
IM-T-006	$\gamma\delta$ T V δ 1 tumor	TRDC, TRGC1, TRGC2, TRDV1	KLRC1, NCR3, CD7, PRF1, GZMA	TRAC-dominant, CD4, CLEC4C, CD3D, CD14
IM-M-001	AS DC / DC5 (AXL+SIGLEC6+)	AXL, SIGLEC6	PPP1R14A, CD22, CD5, SIGLEC1, ITGAX-mid, IRF4	CLEC4C, CD3D, CD14
IM-M-002	pDC	CLEC4C, LILRA4, IL3RA, TCF4, IRF7	GZMB, LILRB4, JCHAIN, PTCRA, MZB1, PLD4	CD1C, XCR1, CD3D/E, NKG2A
IM-M-003	LAMP3+ mregDC	LAMP3, CCR7, BIRC3	FSCN1, CCL19, IDO1, EBI3	CLEC9A, CLEC4C, CD14
IM-M-004	Langerhans (CD207+)	CD207, CD1A	CD1C, EPCAM, RUNX3	CLEC9A, CLEC4C
IM-M-005	TREM2+ LAM (immune-side)	TREM2, APOE, APOC1	GPNMB, CD9, SPP1-mid	CD3D, CD14
IM-M-006	FOLR2+ perivascular	FOLR2, LYVE1	MAF, MRC1, SELENOP	IL1B, CXCL12
IM-M-007	SPP1+ TAM	SPP1, MARCO	MMP9, GPNMB, TREM2-mid	CD3D, CD14
IM-M-008	ISG-M ϕ	IFIT1, IFIT3, ISG15, OAS1	IFI6, MX1, RSAD2, IL1B	FOLR2, TRIM21
IM-B-001	ABC (T-bet+CD11c+)	TBX21, ITGAX, FCRL5	ZEB2, FGR, FCRL3, TNFRSF13B	IGHD-hi, with CD38-hi
IM-B-002	TLS memory B / plasma	MS4A1, CD27, TNFRSF13B, XBP1, MZB1	IGHA1, IGHG1, IGHG3	FCRL5-hi, TIGIT
IM-N-001	KIR+ adaptive NK	KLRC2, BCL11B, IFI44L	ITGAL, ZNF683	FCER1G-hi, CD3E
IM-I-001	ILC1 (tumor)	TBX21, IFNG	KLRC1, XCL1, NCR1-low	NCAM1, CD3E
IM-I-002	ILC2	GATA3, PTGDR2, IL1RL1	RORA, HPGDS, KLRB1	TBX21, CD3D, CD3E
IM-I-003	ILC3 NKp44+	RORC, NCR2, AHR, IL22	IL23R, KLRB1	TBX21, CD3E
IM-G-001	Mast cells (tumor)	TPSAB1, TPSB2, CPA3, KIT, HDC	CMA1, MS4A2, FCER1A	CD3D, CD14
IM-G-002	LOX-1+ PMN-MDSC	OLR1, ARG1, CD84	CXCR4-hi, CEBPB, MMP9	hi, CD68-hi, CD14-hi, CD11b
IM-H-001	CD34+ HSPC contaminant	CD34, KIT-mid, SPINK2	AVP, MEIS1, HOPX, PROM1	MPO-hi, CD3D, CD14
EP-E-001	Pulmonary ionocytes (anchor)	FOXI1, ASCL3, CFTR, ATP6V1G3	BSND, CLCNKB, STAP1	SFTPC, SLC26A9, KRT5, TP63
EP-E-002	Pancreatic epsilon (anchor)	GHRL	ACSL1, ASGR1, BHLHE41, FRZB	INS, GCG, SLC18A3
<i>Tumor-side and boundary populations (co-owned with Tumor Biology, from joint registry SN2 populations.tsv):</i>				
CA-S-001	apCAF (boundary, HLA-DR+ stromal)	HLA-DRA, HLA-DRB1, CD74, SAA3, SLPI	CLU, COL14A1	PTPRC, CD14

con

Table 4 continued

Pop ID	Subset	Positive core	Positive supportive	Exclusion
CA-S-002	myCAF	ACTA2, TAGLN, MYH11, POSTN, MMP11	COL1A1, COL3A1	HLA-DRA, I
CA-S-003	iCAF	IL6, CXCL12, HAS1, HAS2, PDGFRA	CXCL1, CXCL2, LIF	ACTA2, TA
CA-M-001	TREM2+ LAM (tumor-side)	TREM2, APOE, CD9, GPNMB, LGALS3, LPL, FABP5	SPP1, C1QB, CTSD	CD3D, CD19
CA-M-002	FOLR2+ perivascular (tumor-side)	FOLR2, LYVE1, MRC1, SEPP1, CCL18, F13A1	SEPP1, MERTK	CLEC10A, C
CA-V-001	Tip EC	ESM1, DLL4, APLN, PDGFB, NID2, ANGPT2	CXCR4, UNC5B	VWF-high, S
TU-T-001	DTP (drug-tolerant persister)	NR2F1, TXNIP, NGFR, AXL	PROM1, SOX2, JUN, FOS, ATF3	MKI67, PCNA
TU-T-002	EMT-hybrid	VIM, SNAI2, ZEB1, ZEB2, S100A4	CDH2, FAP-low	CDH1-high

Supplementary Note S4 — Immunology calibration set and marker panels

Owner: Immunology (agent-mozus3vv). Content in `agent/immunology:workspace/rare-immune-subsets_catalog.md` and `marker_panels.md`.

Supplementary Note S5 — Wet-lab validation plans for candidate novel motifs

Owner: Immunology. Minimal FACS panel proposals and in-silico replication strategies for selected Kang-atlas motifs receiving the "candidate novel state" call.

Supplementary Note S6 — REAL sensitivity analysis

Owner: Computational Biology + Data Science. Sensitivity of REAL to Leiden resolution, ρ , k , K , μ (witnessing depth), τ , σ_{thresh} , α , and random seed.

Supplementary Note S7 — Runtime and memory

Wall-clock and peak GPU memory for REAL and for each integrator on the Kang 2024 atlas on an 8×A100 node; streaming / approximate variant prescriptions.

Supplementary Note S8 — Cross-atlas replication details

Per-motif reproducibility on HCA Lung Cancer and Gut Cell Atlas tumor arm; Jaccard overlap with main-text motif list.

Table 3: Rare immune and control subpopulations used for REAL detection-framework calibration and RareShield protection benchmarks. Tier 1 = primary stress-test targets; Tier 2 = secondary anchors and confounder sentinels. Markers listed are the highest-specificity positive-core genes; full panels with supportive and exclusion gates are in Supplementary Table S3.

Tier	Subset	Canonical markers (top)	Reference atlas	PMID
1	Pulmonary ionocytes	FOXI1, ASCL3, CFTR	HLCA (Sikkema 2023)	37291214
1	Plasmacytoid DC	CLEC4C, LILRA4, IL3RA	Kang 2024 pan-cancer	38997313
1	AS DC / DC5	AXL, SIGLEC6	Villani 2017 PBMC	28428369
1	MAIT (tissue)	KLRB1, SLC4A10, ZBTB16	HCA Immune (Dominguez 2022)	35549404
1	eTreg (CCR8+TNFRSF9+)	FOXP3, IL2RA, CTLA4, CCR8	Zheng 2021 pan-cancer T	34914499
1	TRM-TR CD103+CD39+	CD69, ITGAE, ENTPD1	Duhen 2018 HNSC	30076360
1	TPEX CXCR5+TCF7+	TCF7, CXCR5, IL7R	Miller 2019 mouse/human	30664895
1	$\gamma\delta$ T V δ 1 tumor	TRDC, TRGC1, TRDV1	Zheng 2021 pan-cancer T	34914499
1	LAMP3+ mregDC	LAMP3, CCR7, BIRC3	Maier 2020 Nature	32269339
1	TREM2+ LAM	TREM2, APOE, APOC1	Jaitin 2019 Cell	31257022
1	FOLR2+ perivascular M ϕ	FOLR2, LYVE1	Nalio Ramos 2022 Cell	35325594
1	SPP1+ TAM	SPP1, MARCO, MMP9	Cheng 2021 pan-cancer myeloid	34004147
1	ABC / atypical memory B	TBX21, ITGAX, FCRL5	Cancro 2020 review	31986068
1	KIR+ adaptive NK	KLRC2, BCL11B	Schlums 2015 Immunity	25786176
1	ILC1 (tumor)	TBX21, IFNG	Dominguez 2022 HCA	35549404
2	T _{FH} /T _{FR} (TLS)	CXCR5, BCL6, PDCD1	Cabrita 2020 Nature	31942071
2	TLS memory B / plasma	MS4A1, CD27, XBP1	Helmink 2020 Nature	31942071
2	Langerhans (CD207+)	CD207, CD1A	Mulder 2021 Immunity	33691135
2	ISG-M ϕ	IFIT1, IFIT3, ISG15	Cheng 2021 pan-cancer myeloid	34004147
2	ILC2	GATA3, PTGDR2, IL1RL1	Dominguez 2022 HCA	35549404
2	ILC3 NKp44+	RORC, NCR2, AHR	Dominguez 2022 HCA	35549404
2	Mast cells (tumor)	TPSAB1, TPSB2, CPA3	Dwyer 2016 Nat Immunol	27019417
2	LOX-1+ PMN-MDSC	OLR1, ARG1, CD84	Condamine 2016 Sci Immunol	27034275
2	CD34+ HSPC contaminant	CD34, KIT, SPINK2	Velten 2017 Nat Cell Biol	28813410
2	Pancreatic epsilon (anchor)	GHRL	Baron 2016 Cell Syst	27667365

Supplementary Note — Unseen-species estimators and the REAL framework

The problem REAL addresses has a long statistical pedigree outside single-cell biology. The *unseen-species problem* — estimating the number of species in a sample when some species are present but unobserved — has been studied since good1953population, who, with Turing, derived estimators during WWII cryptanalytic work. Subsequent developments by chao1984nonparametric, bunge1993species, orlitsky2016optimal and valiant2011estimating produced estimators whose sample complexity is well understood and whose epistemic status is exactly the one we adopt for REAL: a *bound-valued* claim about unobserved quantities, calibrated under exchangeability assumptions that are explicitly named. The purpose of this note is to situate REAL within that tradition, so that reviewers familiar with the species-estimation literature can map our guarantees onto known ground.

The analogue. In the species-estimation setting, one observes frequencies $\{f_1, f_2, \dots\}$ where f_k counts species observed exactly k times in a sample of size n , and asks for the total (observed + unobserved) species count, or the mass of unobserved species. The Good–Turing estimator gives the mass of unobserved species as f_1/n

— the proportion of the sample accounted for by singletons. Chao’s nonparametric estimator gives a lower bound on unobserved species count, $\hat{S}_{\min} = S_{\text{obs}} + f_1^2/(2f_2)$. In both cases, the estimator is *not* a cell-type identification; it is a population-level quantity about *how much mass is invisible*.

REAL occupies the same epistemic slot with one structural difference: where species estimators look at *frequency* of observations, REAL looks at *cross-batch reproducibility of local-density structure*. The analogue of a “singleton species” in REAL is a candidate motif observed in one batch only; the analogue of a “doubleton” is a motif observed in $\mu \geq 2$ batches under our witness threshold. The witness threshold μ is the discretised frequency; the minimax envelope Theorem 2 is the sample-complexity lower bound; the exchangeability bound Lemma 1 is the bridge from what we can count (held-out known rare motifs) to what we want to bound (truly unannotated rare motifs).

What is and is not inherited. The following parallels are load-bearing:

- *Population-level, not individual-level, claims.* Neither Good–Turing nor REAL identifies a specific unobserved species or motif; both estimate *population properties* (total mass, loss rate, detectability envelope). A REAL flag is population-level in this precise sense.
- *Exchangeability as an explicit assumption.* Classical unseen-species estimators are consistent under i.i.d. sampling; any structured nonexchangeability in the sampling distribution (e.g., rare species concentrated in undersampled strata) requires an exchangeability bound analogous to Lemma 1. The philosophy-of-statistics literature on this (e.g., [potochnik2017idealization](#)) treats such assumptions as legitimate idealisations provided they are named and their boundary conditions are stated.
- *Bound, not equality.* Chao’s estimator is explicitly a lower bound on unobserved species, never an equality. REAL’s exchangeability transfer is explicitly an upper bound on unknown-rare loss, never an equality. The bound-valued form is what makes the claim empirically defensible.

The following distinctions are equally important:

- *REAL is not a species count.* REAL outputs a motif set $\mathcal{W}$ of reproducible local-density structures and a per-method survival score $\text{LRS}(f)$. It does not estimate a number of novel cell types; it identifies reproducible structural signals whose erasure under integration should be investigated further.
- *The unit of analysis is a motif, not an individual observation.* Good–Turing operates at the observation level (a single capture of a species); REAL operates at the motif level (a cross-batch-reproducible density signature). This is the appropriate unit for a question about *structural* preservation rather than *frequency* estimation.
- *Hierarchical amplification is not classical.* The telescoping detectability argument of Theorem 9 has no direct unseen-species analogue in the classical literature (though see [orlitsky2016optimal](#) for hierarchical sampling variants). It is a consequence of the compartmental structure of single-cell biology — sub-populations are nested within lineages, not drawn from a flat reference class.

Philosophical reinforcement. The bound-valued stance of [good1953population](#) and [chao1984nonparametric](#) is not a hedge; it is the only epistemically honest form such claims can take. An equality-valued claim about unobserved species, or about unannotated rare motifs, would require observing the thing that is by definition unobserved. The bound-valued form is what classical statistics has always been able to deliver, and it is what REAL delivers. The epistemic-scope paragraph of the Discussion (??) should be read with this heritage in view: nothing about REAL’s bounds is novel as a philosophical commitment; what is novel is the mapping of the commitment onto the structure of single-cell batch integration.

Further reading. [valiant2011estimating](#) gives sample-complexity-optimal estimators for unseen quantities; their analysis techniques map onto the Le Cam argument of Theorem 2. [bunge1993species](#) surveys the species-estimation literature through 1993. [orlitsky2016optimal](#) extends Good–Turing to the sample-complexity-optimal regime.

Cancer-genomic and immunologic applications. In oncology specifically, Chao-style unseen-species estimators have been applied since the mid-2010s to two distinct problems directly cognate to REAL’s rare-subpopulation question. First, to estimate unsampled tumour *subclones* from observed variant-allele-frequency (VAF) distributions: williams2016identification and sottoriva2015big use VAF-diversity indices built on Chao-like reasoning to infer subclonal heterogeneity beyond direct sample coverage, and the TRACERx programme extended the approach to multi-region NSCLC debruin2014spatial,jamalhanjani2017tracking. Second, to estimate unsampled *T- and B-cell receptor clonotypes* from observed tumor-infiltrating repertoires: valpione2020immune (melanoma) and reuben2020comprehensive (NSCLC) apply Chao1 and Chao–Shen-corrected Shannon indices to the diversity of observed TIL repertoires to estimate what the repertoire would look like under complete sampling. In both settings the claim-shape is exactly the one we adopt: a bound-valued, population-level, bounds-not-equality statement about what the data did not directly sample, calibrated under stated exchangeability assumptions. REAL extends this tradition from allele counts and receptor clones to transcriptomic rare-subpopulation detection, using cross-batch reproducibility in the HVG space as the analogue of cross-sample detection in allele or receptor space.

Supplementary Note S9 — Ethics and data use

All data is publicly available; no new human-subjects work is reported.

Supplementary Table S1 — Datasets

(See Note S2.)

Supplementary Table S2 — Rare immune subsets and canonical markers

Placeholder; content from `rare-immune-subsets-catalog.md`.

Supplementary Table S3 — Hyperparameters and seeds

Full sheet of every integrator version, every REAL / RareShield hyperparameter value, every random seed used in the paper.

Author contributions (full)

- **Biological Science** (`agent-mozuoqax`): scientific framing, manuscript integration, writing, cover letter, research-rigor review gate.
- **Data Science** (`agent-mozutvsv`): REAL framework design, figure plan and all figures, benchmark harness.
- **Computational Biology** (`agent-mozur9ik`): implementation, GPU experiments, scaling.
- **Mathematics** (`agent-mozusggu`): formal detectability theorem, identifiability, null calibration.
- **Machine Learning** (`agent-mozurrdm`): literature positioning, RareShield loss design.
- **Immunology** (`agent-mozus3vv`): immunological framing, rare-subset catalog, MAIT/eTreg/pDC experiments, motif interpretation.
- **Tumor Biology** (`agent-mozurh1r`): oncological framing, rare cancer subpopulation registry, Kang motif cancer-type interpretation, clinical-implications discussion.
- **Philosophy** (`agent-mozur8ub`): epistemic framing, limitations subsection, hidden-assumption audit.